

Benjamin Flanders

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Experience

Artimatic Technologies, Inc.

Athens, GA

Technical Co-Founder

January 2020–Present

- Implemented AI-based solutions and custom machine learning models using PyTorch and TensorFlow; built and maintained SaaS products with NextJS, TypeScript, React, and AWS (SageMaker, EC2, Amplify).
- Managed full-stack development, including RESTful API backend, database architecture, and multi-site deployment, ensuring robust and scalable solutions.
- Handled all aspects of bookkeeping, grant accounting, and board of directors planning.
- Cultivated advisor and investor relationships.
- Secured over \$300k in grant and loan financing.

Wayfinder Data Solutions, LLC.

Atlanta, GA

Freelance Programmer

January 2023–Present

- Designed and developed a multi-user data entry and analytics web app supporting multiple construction companies, improving project tracking accuracy and reducing project runtime by over 5%.
- Managed project lifecycle using industry-standard software management tools and Agile methodologies, ensuring timely delivery and client satisfaction.
- Handled invoicing, payments, and client communication to prioritize and deliver essential features.

National Parks System

Athens, GA

Graduate Assistantship / Database Programmer

Aug 2019–May 2021

- Developed and maintained a VBA-based data management application and a Transact SQL database for the NPS SECN Estuarine Water Quality Team
- Developed an audio data management application and database for the NPS SECN Wildlife Team using Visual Basic and Transact SQL.
- Authored and published database and application documentation.

Kids 4 Coding

Roswell, GA

Technology Instructor / Curriculum Developer

May 2017–December 2017

- Taught over 120 4th - 8th graders computer science fundamentals, programming, game development, project management, and team building skills
- Collaborated with other teachers to provide a better student experience.

Education

The University of Georgia, Athens

Athens, GA

MS in Artificial Intelligence 3.7/4.0 GPA

Aug 2019–May 2021

Coursework: : Artificial Intelligence, Human Computer Interaction, Data Mining

Activities: : Master's Thesis, Graduate Assistantship

The University of Georgia, Athens

Athens, GA

BS in Computer Science, BA in Cognitive Science 3.4/4.0 GPA

Aug 2015–May 2019

Coursework: : Software Engineering, Introductory Statistics, Linear Algebra

Dean's List: Spring Semester of 2017, Fall Semester of 2017, Spring Semester of 2018

Projects

Thesis: Machine Learning Classification Of National Parks Bird Audio Data

Dec 2019–Apr 2021

- Produced a machine learning classifier capable of segmentation and species classification of bird vocalization audio data
- Authored and defended a thesis documenting a novel machine learning process and related publications
- Designed and implemented a novel human in the loop methodology

Patents

Automated weighting generation for three-dimensional models

Patent No. 11995771

Inventor

June 2022

- Developed and patented a novel process that creates and adjusts model weightings matrices, applies a machine learning model, and generates an accurate output weighting.
- Collaborated with a team to develop and author the patent.

Publications

World Academy of Science, Engineering and Technology

Oct 2023

Automated Weight Painting: Using Deep Neural Networks to Adjust 3D Mesh Skeletal Weights

Grants

National Science Foundation - SBIR Phase 1 Grant

Atlanta, GA

Principal Investigator

January 2024–December 2024

- Led proposal authorship.
- managed grant accounting, hiring, and strategic planning.
- Developed and integrated multiple interworking ML models, each serving distinct functions, to provide comprehensive animation support to customers. Leveraging graph neural networks, sparse matrix representations, generative neural networks, and transformers.
- Built a Python library capable of processing GLB files into Machine-Learning ready datatypes.
- Designed and implemented cloud pipelines for ingesting

Public Speaking

Film and Media Exchange Conference

Stuttgart, Germany

Speaker

May 2022

- Led a presentation on the relevance and application of Machine Learning to the Animation Industry and how to generally apply ML to animation projects.

Skills

Programming Languages: Python, TypeScript

Web Development: NextJS, React, Tailwind CSS

Cloud Services: AWS (including Lambda, EC2, SageMaker)

Machine Learning Frameworks: PyTorch, TensorFlow, Hugging Face

Version Control: Git (GitHub)

Project Management: Agile methodologies

Financial Management: QuickBooks

Grant Management: Grant accounting, proposal writing

Team Management: Hiring, managing developers

Other Skills: AI/ML model development, SaaS product development, RESTful API development